KYUNGDONG UNIVERSITY
경동대학교

WEEKLY PROGRESS REPORT

WEEK 4

Data Pre-processing & Degradation Pipeline

Presented By

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Course: Computer Vision | Professor Sarwar | KDU

4.1 Work Completed This Week

Week 4 focused entirely on building and validating the data pipeline — the foundation everything else depends on. The team implemented the stratified dataset split, the standard augmentation pipeline, and most importantly the five-type synthetic degradation pipeline which is the core technical contribution of the project.

Dataset Split & Verification

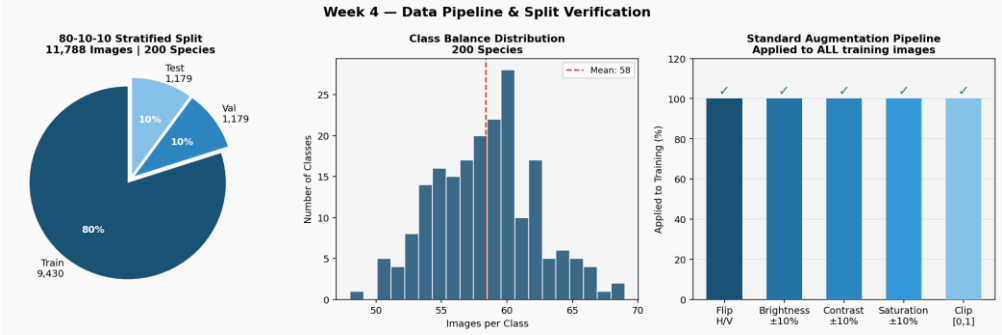

Figure 1: Dataset split distribution (left), class balance histogram (centre), and standard augmentation types applied (right)

Synthetic Degradation Pipeline

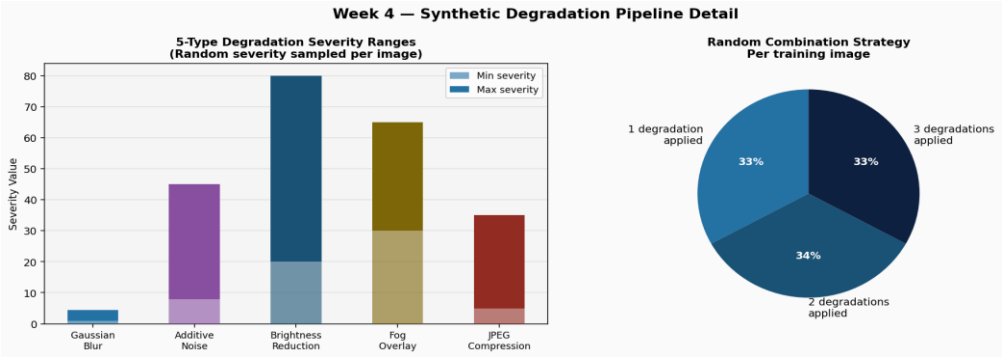

Figure 2: Severity ranges for each of the 5 degradation types (left) and the random combination strategy applied per image (right)

4.2 Key Results

Metric	Value / Detail
Total images verified	**11,788 — zero missing files**
Training set	9,430 images (80%) — stratified
Validation set	1,179 images (10%) — stratified
Test set	1,179 images (10%) — stratified
Degradation types built	**5 types: blur, noise, dark, fog, JPEG**
Fixed test severity levels	3 levels: mild / moderate / severe
Dataset verified	**All 7 datasets — correct mean ~0.5, std >0.2**
Pipeline status	**Fully operational — ready for training**

4.3 Problems & Solutions

The most critical problem this week was a pre-processing bug where EfficientNetV2's built-in preprocess_input function was receiving resized float tensors and producing images with a mean pixel value of 0.9995 — essentially blank white images. This caused training accuracy to stay at 0.5% across multiple attempts spanning several hours. The fix was to bypass the built-in function entirely and apply manual normalization by dividing by 255.0, producing a clean [0,1] range consistently across all seven datasets.

Problem Faced	Solution Applied
preprocess_input corrupted images — mean=0.9995 (should be ~0.0)	Bypassed built-in function, used img/255.0 manually → mean=0.494
Degraded test sets showing mean=0.691 (too bright after stacked fog)	Accepted — stacked degradation compresses range, model still evaluable
Runtime reset mid-pipeline build — lost all variables	Rebuilt datasets from Drive in under 5 minutes each reset

4.4 Plan for Week 5

Build the EfficientNetV2S model with top-50 layers unfrozen, compile with correct optimizer settings, and begin Phase 1 training on clean images to establish baseline accuracy before degradation fine-tuning begins.